

Name: _____ Date: _____

Period: _____

Exit Ticket — Answer Key

1. Given $\sin \theta = \frac{\sqrt{3}}{2}$ and $\cos \theta = \frac{1}{2}$, compute $\tan \theta$. Show your work.

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3}$$

2. State the period and the location of the two positive vertical asymptotes closest to $\theta = 0$ for the function $g(\theta) = \tan(2\theta)$.

Period of g : $\pi/2$

Asymptotes: $\theta = \pi/4$ and $\theta = 3\pi/4$

Teacher Note

Asymptotes of $\tan(2\theta)$: set $2\theta = \pi/2 + k\pi \Rightarrow \theta = \pi/4 + k\pi/2$. The two smallest positive values are $\pi/4$ ($k = 0$) and $3\pi/4$ ($k = 1$).

3. For the parent function $y = \tan \theta$, the graph is always **increasing** between consecutive asymptotes, and changes from concave **down** to concave **up** within each period.